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Xiang (Shawn) Li, Ph.D.

INTERESTS

Computational Biology | Scientific AI Platforms | Cloud & HPC

EDUCATION

Ph.D., George Washington University, Physics (**Honor: Parke Prize**), 2016 ~ 2021
Thesis Summary Study of Transcriptional and Post-transcriptional Regulation in Immune Systems
B.S., Huazhong University of Science & Technology, Applied Physics, (**Honor**) 2011 ~ 2015

SKILLS & Tools: AWS CCP, DeepLearning.AI, Nextflow,

- **Agentic AI:** LangChain, LangGraph, Model Context Protocol (MCP), mem0, Harbor
- **Deep Learning** Tensorflow, PyTorch, Deeplearning4j, AlphaFold2, AlphaGenome,
- **Programming :** Python (Data mining → Visual → ML), R, C/C++, MATLAB, Java, Fortran, PyPI, LaTeX
- **Bioinformatics & Workflow Engines:** Nextflow, WDL, Snakemake, Flask, Dash, Streamlit,
- **Cloud & DevOps:** Linux, AWS (S3, EC2), Docker, PostgreSQL, Caddy, Nginx
- **Biological Databases:** gnomAD, HuBMAP, SenNet, BIL, GEO, ENCODE, HapMap, dbSNP

Professional Experiences

PSC, Carnegie Mellon University Senior Scientist — Pittsburgh PA, 2024 ~ Present

- **Partnership** Human BioMolecular Atlas Program: Led the collaboration between TACC & PSU Galaxy, By building AI-enabled research infrastructure and workflows(HuBMAP) under (CFDE)
- **Platform Built**: AI-assisted data research + multi-agent analytics infrastructure for HuBMAP Data Portal.
- **Support**: Contributors to Open-source AI projects: Harbor, LangChain & Internal Evaluation
- **Principal Investigator** (NIH R03 RFA-RM-26-017): Architecting a multi-agent scientific platform (using MCP server + LangGraph) to automate cross-repository scientific queryting, evidence synthesis and bioinformatic workflow execution.

Johnson & Johnson Scientist III, RWE — Titusville NJ, 2024

- Engineered an end-to-end Python analytical toolkit from scratch to ingest, align, and visualize longitudinal patient trajectories across large-scale EHR/EMR datasets for clinical cohort profiling.
- Profiling 53K patients for alignment, visualization and analysis.

GentiBio, Senior Scientist, Bioinformatics Lead — Boston MA, 2022 ~ 2024

- **Translational Research**: GNTI-122 IND package (Safety assessment for Gene Editing).
  - *Build SNP-aware (gnomAD v4.0) insilico Off-target analysis for Cas9 editing
  - *On/Off-Target analysis protocol: in-Silico ⇒GUIDESeq ⇒rhAMPSeq.
  - *Build a full-stack web app (AWS) for automation. Similar to CRISPResso2 Web
- **Early Discovery**, Scientist II: Cellular Profiling Platform
  - Research: single cell multi-omics integration analysis
  - Dev: AWS VC-backed start-up package and build bioinformatics infrastructure

Sana Biotechnology Scientist, Computational Biology, — Seattle WA, 2021 ~ 2022

- **Corporate team**, research on target discovery & dev pipelines automation.
- **iPSC Master Cell Bank**, Cell Characterization & Safety Assessment

GWU, Department of Physics, Washington, DC, 2017 ~ 2021
Ph.D, (Supervisor: *Prof. W. Peng* , Collaborator: *Prof. H. Xue*)

- (*Nature Com* & JEM), Investigate cis-regulatory roles of Tcf1/Lef1 in $CD8^+$ T cells.
- Integration of Multi-omics: scRNA-seq, RNA-seq and DNase-seq, ATAC-seq, ChIP-seq, CUT&RUN, Hi-C, MiCro-C, HiChIP. Develop the tool HiCHub for 3D genome-based integration.
- Nucleic Acids Research: mRNA stability Mathematical Modeling and data analysis.
- **Tech**: Build analysis pipeline, Cloud & HPC (Linux Admin, data storage & sharing & backup, computation server setup and on-call for tech emergency)
- **Teaching:** Instructor for General Physics (with: *X. Qiu* / *G. Younes* / *S. Guiriec*/ *Kalvir Dhuga*)

Membership & Reviewer

- Phenomics, Editorial Board Member 2024 ~ Present
- Society of Artificial Intelligence Research SAIR 2023 ~ Present
- Machine Learning for Genomics MLGenX 2024

Honors and Awards

- Best Posters Award, Common Fund Data Ecosystem (CFDE Spring 2026)
- The Parke Prize, awarded for excellence in theoretical biophysics (GWU, 2022)
- Project Awards: Silver; International Biomolecular Design Competition 2014, *BIOMOD, Harvard*
- Fellowship $12000 total, Undergraduate Fellowship from College of Life Science & Technology,

Selected Publications & Patents & Presentations

- HuBMAP Data Portal: A Resource for Multi-Modal Spatial and Single-Cell Data of Healthy Human Tissues Morgan T., ..., **Xiang Li**, ..., and Nils Gehlenborg†., **Nature Communications (2026)**
- Inhibition of LARP4-mediated quiescence exit of naive CD4+ T cells ameliorates autoimmune and allergic diseases Jian Zhou, ..., **Xiang Li**, ..., Yi Tian†., **Nature Biomedical Engineering (2025)**
- HiCHub: A Network-Based Approach to Identify Domains of Differential Interactions from 3D Genome Data **Xiang Li**†, ..., Hai-Hui Xue, Weiqun Peng., Bioinformatics (BIOINF-2025-0207)
- Tcf1 CTCF cooperativity shapes genomic architecture to promote CD8+ T cell homeostasis Q. S†, ..., **Xiang Li**,..., Hai-Hui Xue., **Nature Immunology (2022)**
- Tcf1 and Lef1 maintain CD8+ T cell identity by organizing genomic architecture Q. S†, **Xiang Li**†,..., Hai-Hui Xue., **Nature Communications (2021)**
- Sepsis leads to lasting changes in phenotype and function of memory CD8 T cells Isaac J. Jensen†, **Xiang Li**†,..., Weiqun Peng and Vladimir P. Badovinac., 2021 eLife
- Transcriptome-wide stability analysis uncovers LARP4-mediated NFB1 mRNA stabilization during T cell activation. Tian Y.†, Zeng Z.†, **Xiang Li**†, ..., Peng W. and Zhu J. Nucleic Acids Research (2020).
- **Selected Oral Presentation:** 2021 Network Biology, CSHL; Network Study Reveal the critical role of TCF1/LEF1,